

Can Audio LLMs Understand Prosodic Focus?

An Alternative-Semantics Inference Test with *Only*

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“[With writing,] you give your disciples not truth, but only the semblance of truth ... only ... orally ... is there clearness and perfection.”

— Plato, *Phaedrus*, ca. 370 BCE

Overview

- A new inference test where the **correct answer depends on where the focal accent falls** – information that is lost when audio is transcribed to text
- Grounded in **alternative semantics** (Rooth 1992): focus introduces alternatives, and *only* rules them out
- Four native audio LLMs range from **chance to ~90%** accuracy – models struggle to identify focus and draw the correct inference based on it

Association with Focus

Accent placement systematically changes truth conditions:

- Sam only gave **SUE** oranges.
⇒ Sam gave oranges to *no one else*.
- Sam only gave Sue **ORANGES**.
⇒ Sam gave Sue *nothing else*.

Focus introduces a set of **alternatives** (*gave x oranges vs. gave Sue y*); *only* asserts that all of them are false. So *Sam also gave Mary oranges* contradicts (1) but is compatible with (2).

The Task

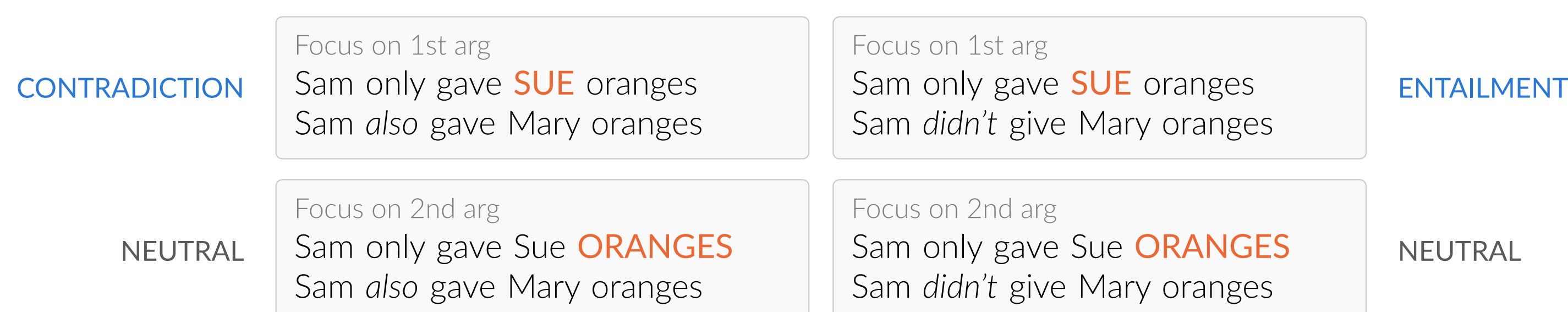


Figure 1. The location of focus determines the correct NLI label.

A 2×2×2 design

- Focus:** 1st argument (person) or 2nd argument (fruit) in sentence 1
- Alternative:** which argument differs in sentence 2
- Logic:** sentence 2 is positive (*also*) or negative (*didn't*)

Each pair of cases differs *only* in focus position, and focus flips the label (ENT/CON ↔ NEU). A model ignoring audio can't do better than **50%**; GPT-5 given text only scores **49.3%**.

Data and tasks

- 128 sentence pairs (16 per structure), recorded by a native U.S. English speaker (Speaker 0); 24-item subset recorded by two additional speakers
- Inference:** 3-way classification (ENT / NEU / CON), with free-text explanation
- Focus ID:** transcribe sentence 1, marking the focused word in UPPERCASE

Probing Open-Weight Models

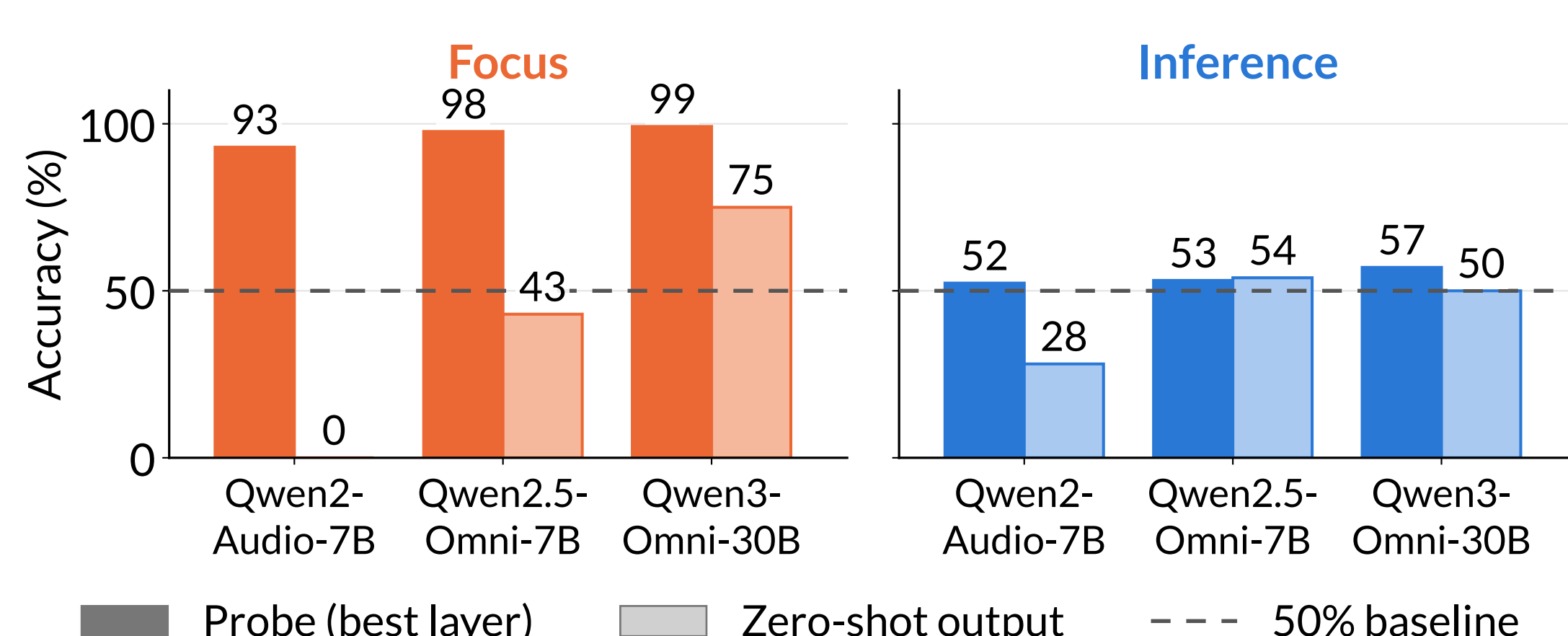


Figure 2. Best-layer linear probes vs. zero-shot output. Focus is linearly decodable from internal states; inference is not. Dashed line: chance for focus, text-only baseline for inference.

Results

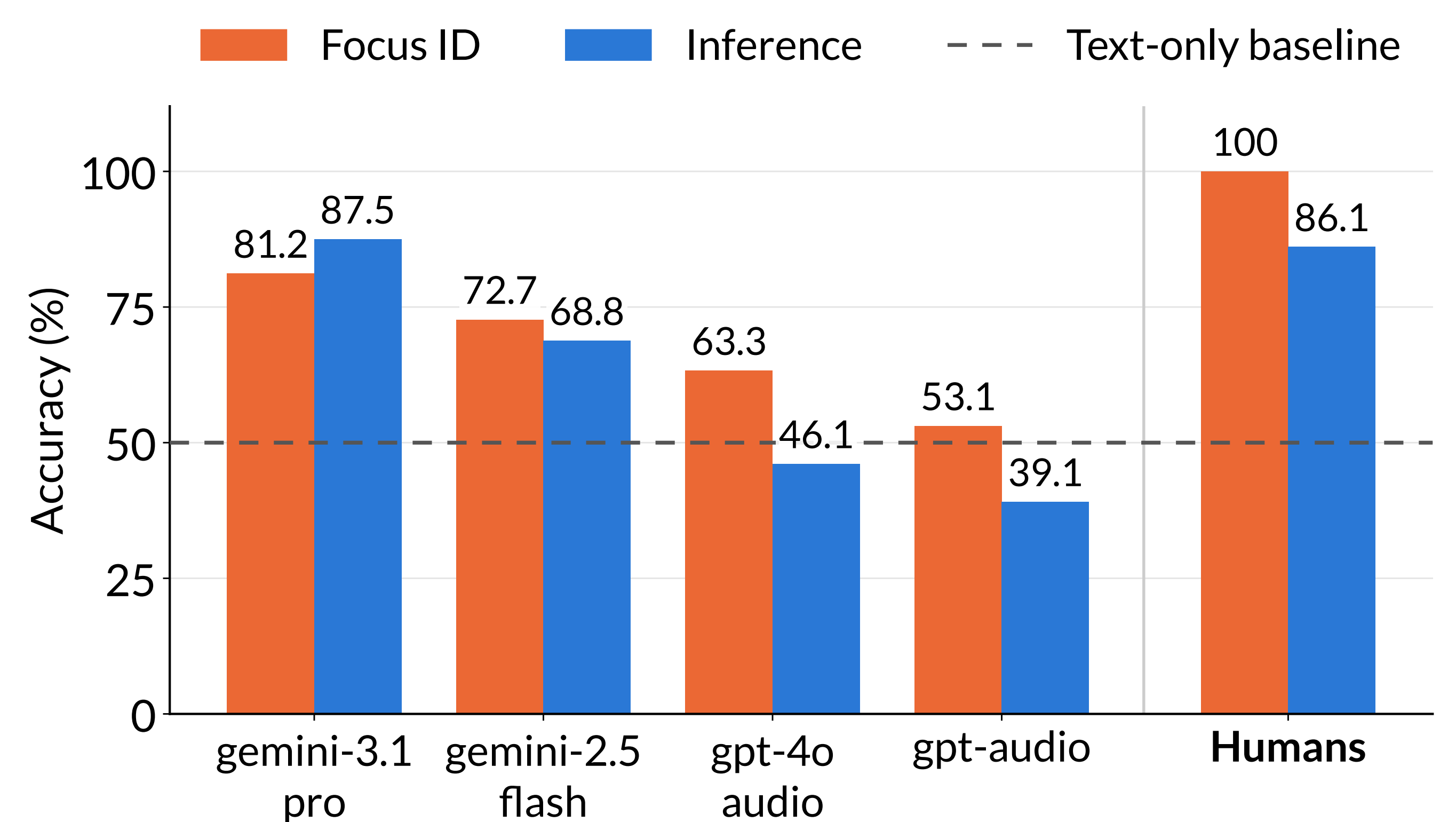


Figure 3. Inference and focus ID accuracy, Speaker 0, zero-shot (N = 128); Humans: online survey of native English speakers (3 respondents, 24-item subset).

- For gemini-3.1-pro, inference task is correct more often than Focus ID task (87.5 vs. 81.2; see Fig. 4)
- Few-shot demonstrations (2 or 5 examples) gave no significant improvement; here we report zero-shot
- Performance is **speaker-sensitive**: on Speaker 2, whose intensity range is much narrower, gemini-3.1-pro's inference accuracy drops from 95.8% to 54.2% on the 24-item subset ($p = .002$)

Task Result vs. Explanation

- Models produce inference task explanations
- An LLM judges whether each explanation gets focus identification and inference right
- Task and explanation provide different views on what model understands – for gemini-3.1-pro, focus explanation is always correct (112/112) when inference task is correct

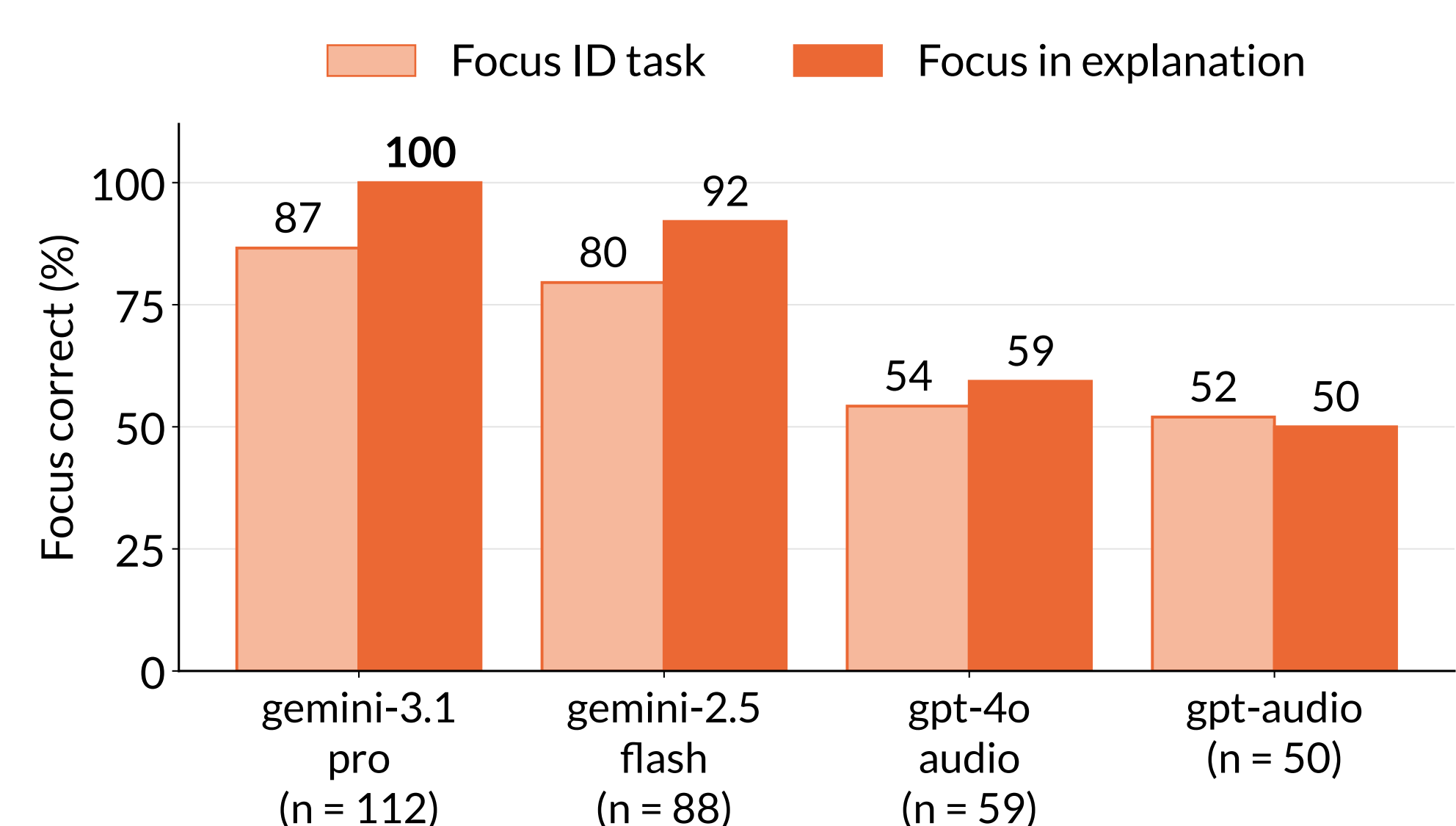


Figure 4. Among items where the inference is correct: how often is focus correct in the Focus ID task vs. in the model's explanation? (Speaker 0, explanations judged by an LLM)

Takeaways

- It is spoken language that is fundamental – written language is “an optional accessory” (Pinker 1994); LLMs were built the other way around, text first
- The current results show that both identifying focus and interpreting its meaning are challenging for current audio models
- Alternative semantics offers a systematic framework for testing audio models in many settings, including other focus-sensitive adverbs, question–answer congruence, and contrast



Code, data and paper:

github.com/DanHardtDK/audio-focus-inference-research